Analysing Data

Week 4, Franziska Pannach

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faculteit der letteren

Disclaimer: Some of the slides in this presentation are taken or adapted from slides from previous AD classes by Xiaoyan Yang and Tommaso Caselli.

Recap

So far, you learned about …

- Motivation for text pre-processing
- Relevant pre-processing steps and approaches
- Tokenization, Stemming, Lemmatization, POS-tagging
- Named Entity Annotation and Recognition
- Evaluating manual and automatic annotations
- Language models
- Embeddings & vectorizing texts using TF-IDF
- Topic Modelling

Recap

Think-Pair-Share

Prepare a 1 minute **explanation** or **answer** to the content/question on the following slides. The slides can also be found on Brightspace (Week 4).

10:00 mins

TF-IDF: Term Frequency-Inverse Document Frequency

- Document frequency: in how many document a word appears
- **Inverse** document frequency: give words that appear in less documents higher importance (salience)

$$\mathrm{tf}_{t,d} = \begin{cases} 1 + \log_{10} \mathrm{count}(t,d) & \text{if } \mathrm{count}(t,d) > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\mathrm{idf}_t = \log_{10}\left(\frac{N}{\mathrm{df}_t}\right)$$

$$\boxed{w_{t,d} = \mathrm{tf}_{t,d} \times \mathrm{idf}_t}$$

Word Embeddings

"You shall know a word by the company it keeps" (John Firth)

	aardvark	...	computer	data	result	pie	sugar	...
cherry	0	...	2	8	9	442	25	...
strawberry	0	...	0	0	1	60	19	...
digital	0	...	1670	1683	85	5	4	...
information	0	...	3325	3982	378	5	13	...

Figure 6.6 Co-occurrence vectors for four words in the Wikipedia corpus, showing six of the dimensions (hand-picked for pedagogical purposes). The vector for *digital* is outlined in red. Note that a real vector would have vastly more dimensions and thus be much sparser.

Maximum Likelihood Estimation

With enough data, you get a pretty good idea of common patterns in a language:

	i	want	to	eat	chinese	food	lunch	spend
i	5	827	0	9	0	0	0	2
want	2	0	608	1	6	6	5	1
to	2	0	4	686	2	0	6	211
eat	0	0	2	0	16	2	42	0
chinese	1	0	0	0	0	82	1	0
food	15	0	15	0	1	4	0	0
lunch	2	0	0	0	0	1	0	0
spend	1	0	1	0	0	0	0	0

Figure 3.1 Bigram counts for eight of the words (out of $V = 1446$) in the Berkeley Restaurant Project corpus of 9332 sentences. Zero counts are in gray. Each cell shows the count of the column label word following the row label word. Thus the cell in row **i** and column **want** means that **want** followed **i** 827 times in the corpus.

Inter-Annotator Agreement

What is inter-annotator agreement and what does a low inter-annotator agreement mean for the outcomes of an annotation task?

Table 2.3: Landis and Koch [9] interpretation of the strength of agreement

Value	Strength of Agreement
< 0.0	Poor
0.00-0.20	Slight
0.21-0.40	Fair
0.41-0.60	Moderate
0.61-0.80	Substantial
0.81-1.00	(Almost) perfect

Recap

Performance Measures for Automated Annotation

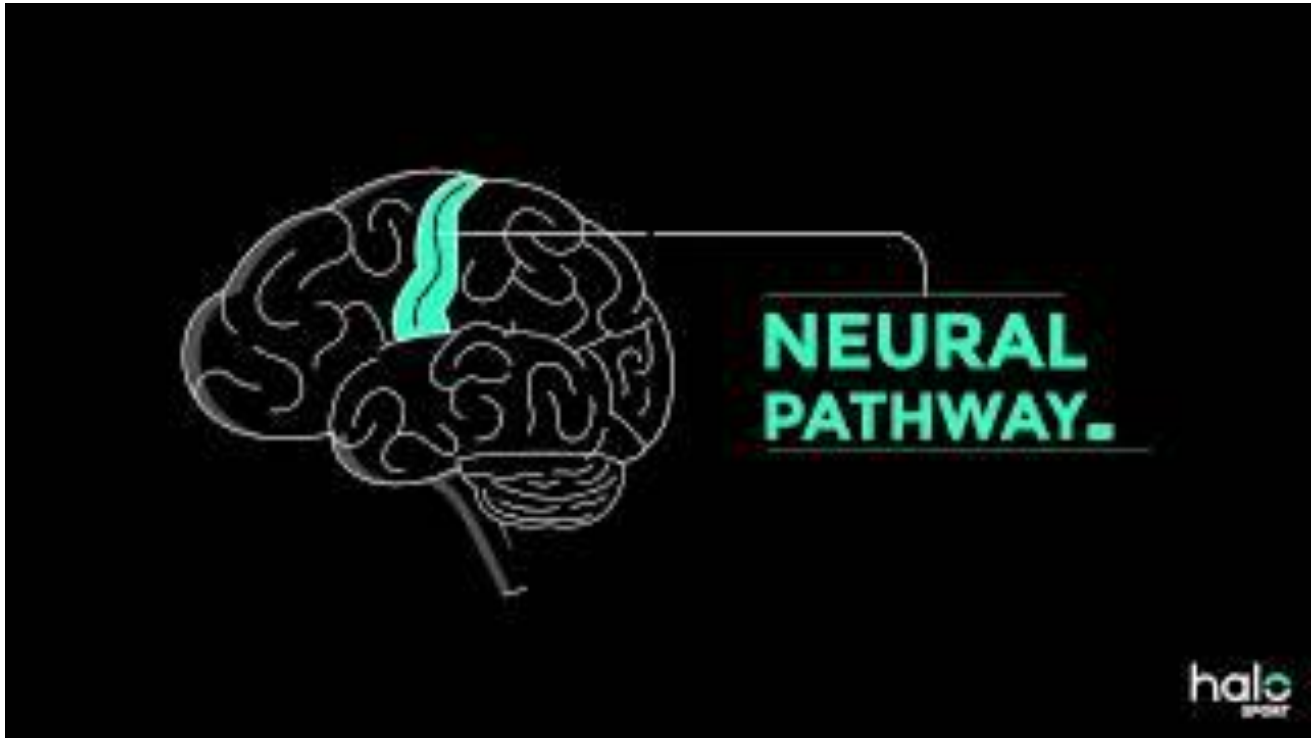
What is the motivation behind Precision, Recall and F1-Score?

		Actual +	Actual -
Predicted	Y	True positives	False positives
	N	False negatives	True negatives

After this week, …

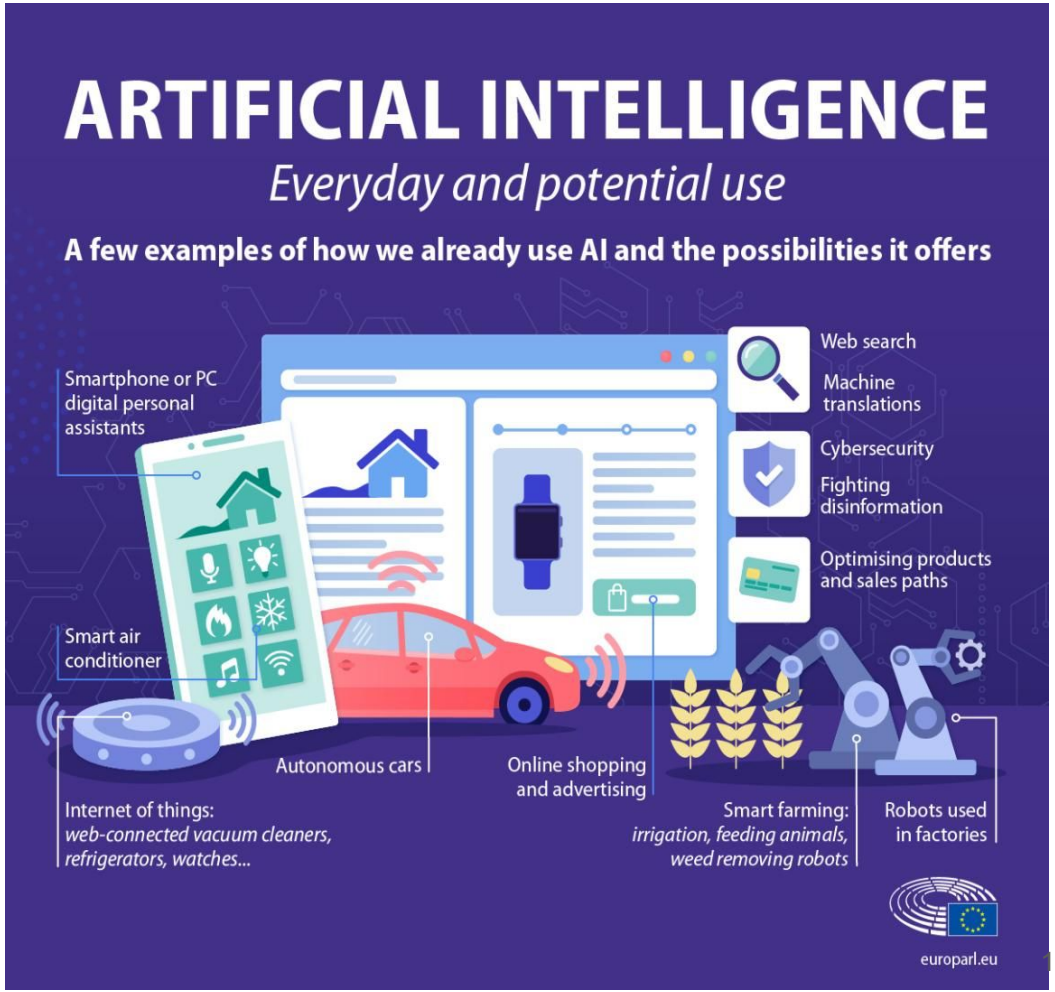
- You will know what machine learning is
- You will get to know typical research questions in DH that use ML
- You will understand what *Linear* and *Multiple Linear Regression* are
- You will know how to apply linear regression to humanities data

What is ~~Machine~~ Learning?



Artificial Intelligence?

- Software or embodied
- “Agents” can be trained models or heuristics!
- AI is an **old** subdiscipline of Computer Science



What is Machine Learning?

- Enables a system to autonomously learn from given information
- Improve without being explicitly programmed (heuristics)

= training

What is Machine Learning?

- Enables a system to autonomously learn from given information
- Improve without being explicitly programmed (heuristics)

  = **training**

- Machine learning algorithms recognize patterns in the **training** data
- Make predictions when confronted with new data

What is Machine Learning?

- Machine learning is usually divided into three categories:
  1. Supervised ML
  2. Unsupervised ML
  3. Reinforcement Learning

Supervised Machine Learning

- Class of machine learning approaches
- Use **labeled** training data to map specific input to specific output
- Training data = Gold Standard, e.g. manually annotated

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Common supervised learning algorithms:

Linear regression

K-nearest neighbors

Naive Bayes

Decision trees

Unsupervised Machine Learning

- Class of machine learning approaches
- Training data not labelled
- Algorithms learn **patterns** from data
- Descriptive modelling and pattern recognition/matching

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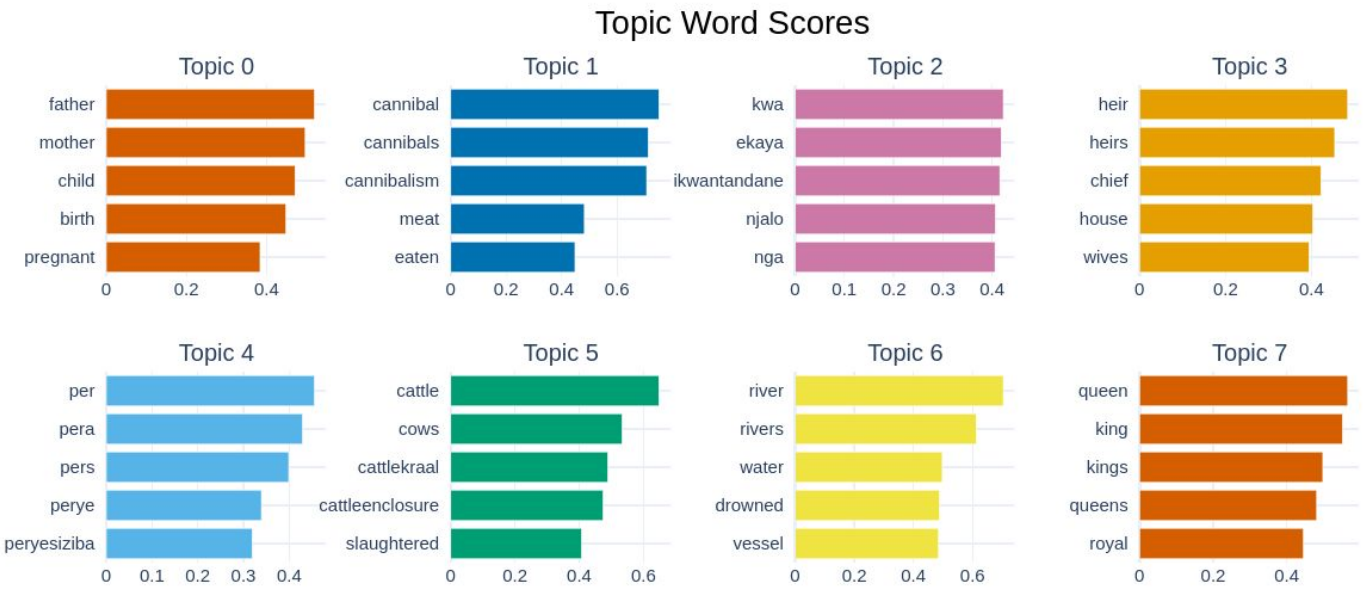
Topic Modelling with BERTopic: Case Study



Unsupervised Machine Learning

- Class of machine learning approaches
- Training data not labelled
- Algorithms learn **patterns** from data
- Descriptive modelling and pattern recognition/matching

Common unsupervised learning algorithms:

K-Means (Clustering)

Principal Component Analysis, PCA (Feature extraction)

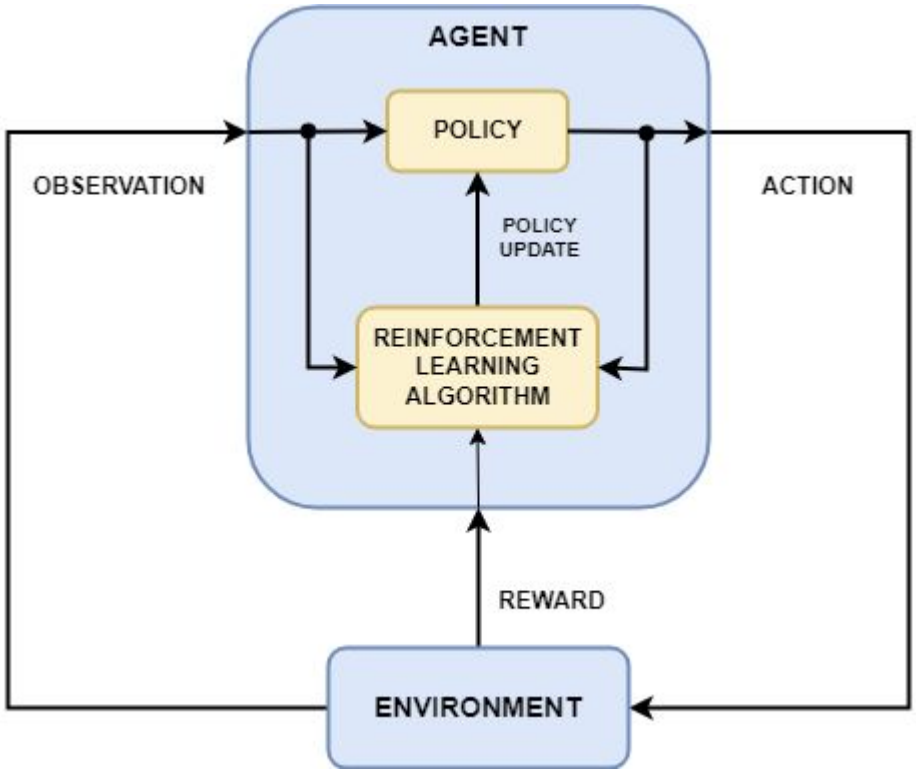
Association rule learning, ARL (Pattern mining)

Reinforcement Learning

- “Learning by doing” through trial and error experiments
- “Agent” learns to perform a defined task through a feedback loop until its performance is within a desirable range
- Positive reinforcement when it performs the task well
- Negative reinforcement when it performs poorly

Reinforcement Learning

- Agent initially makes (random) choices
- Outcome is observed, **rewarded** or **punished**
- After each iteration, policy (=weights) is updated
- Training stops when agent Performs well enough or max. number of iterations are reached



Source: MathWorks

HOW TO SHOOT A FREE THROW
Supervised, unsupervised or reinforcement?
0:39 / 3:00 • Muscle Memory
halo
SPORT

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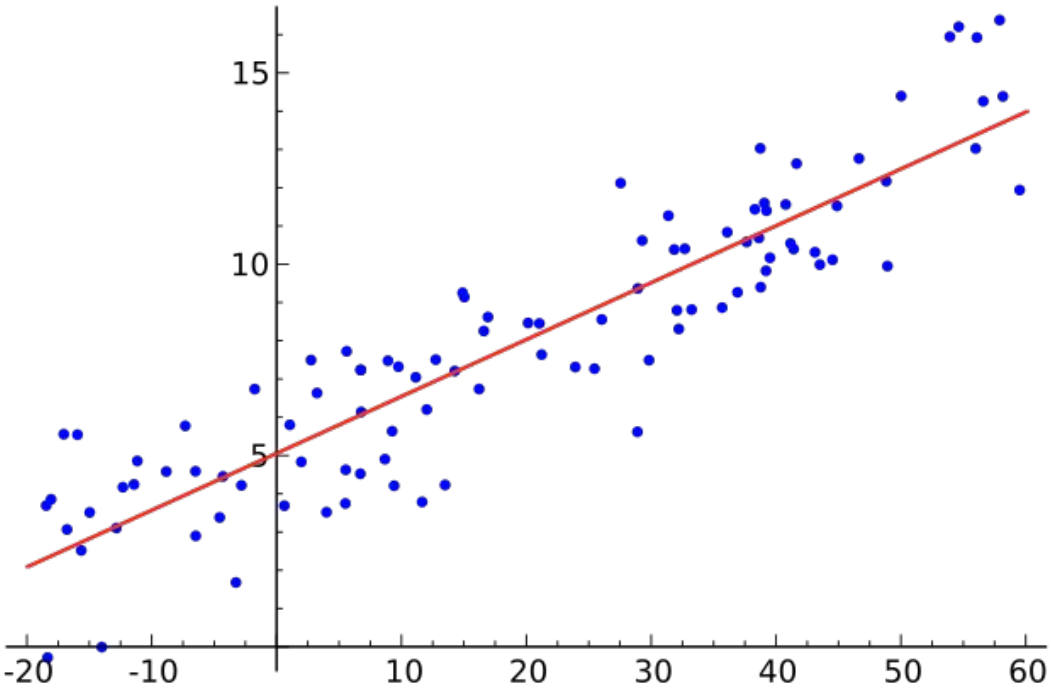
Supervised, unsupervised or **reinforcement**?
0:39 / 3:00 • Muscle Memory
halo SPORTS

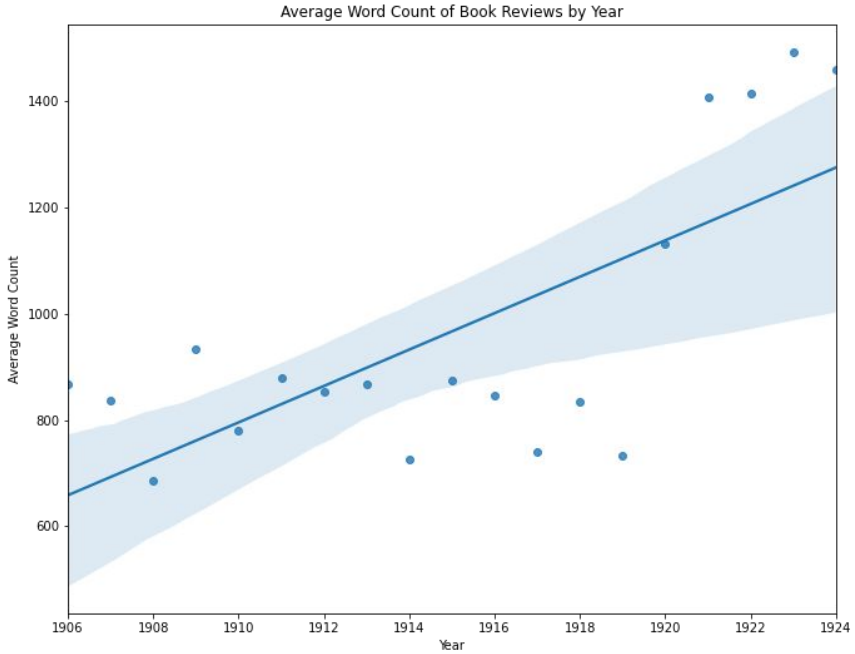
Today: Linear Regression



Source: Wikimedia Commons

Linear Regression

- one **continuous**, independent variable
- one **continuous**, dependent variable



Source: Matthew Lavin, [Programming Historian]

Linear Regression

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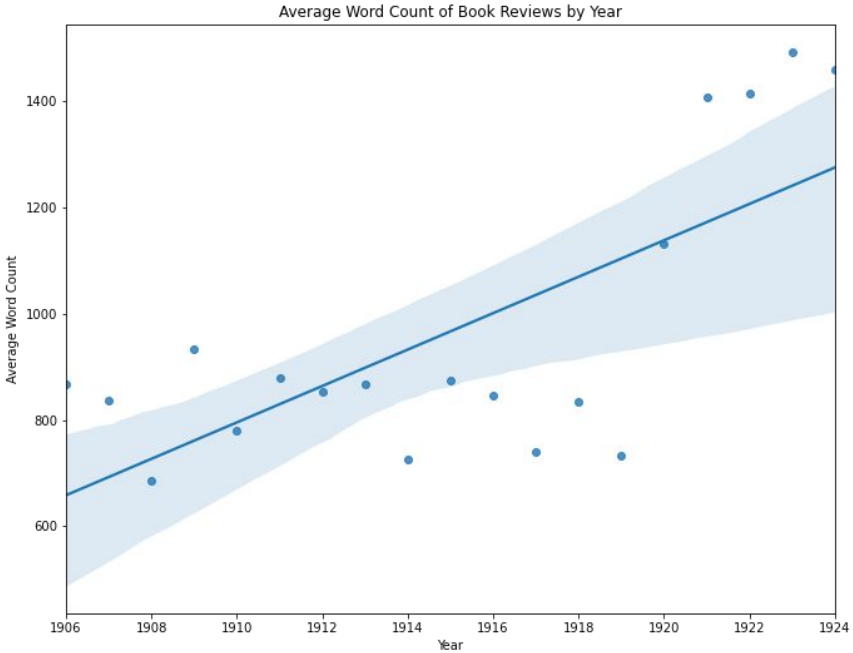
- one continuous, **dependent** variable



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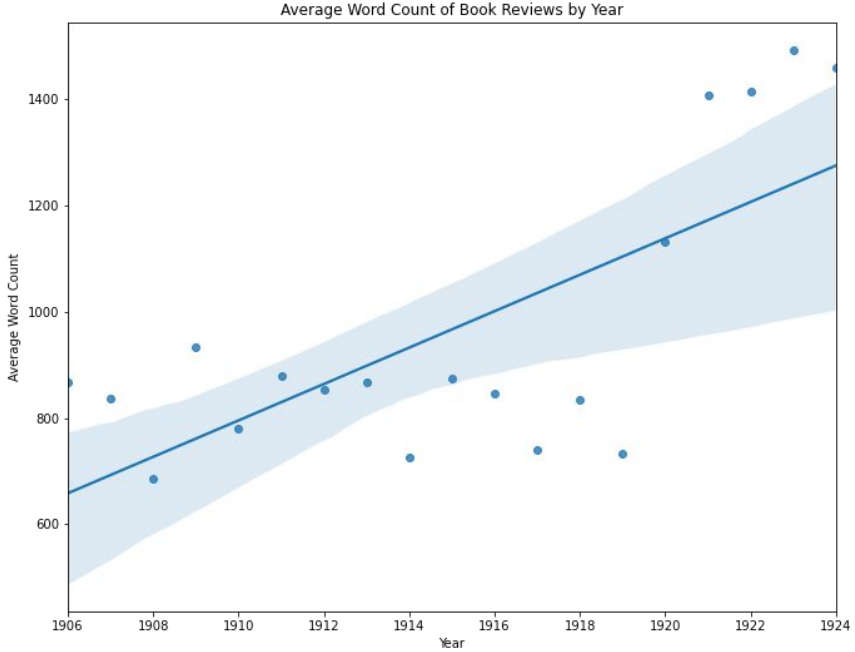
Check **Collecting Data Week 7!**

Dependent and independent variable

- **Independent variable:**
  - The one you manipulate, change, categorize, etc.
  - Often seen as "cause"
- **Dependent variable:**
  - The one you measure, observe, etc.
  - Often see as "effect"
- **Note:** in reality, the relationship may not be causal!

Linear Regression

The goal: a **best fit linear function** (= regression) for given data and evaluate on the **residual,** i.e. how far away data points are from the regression line.



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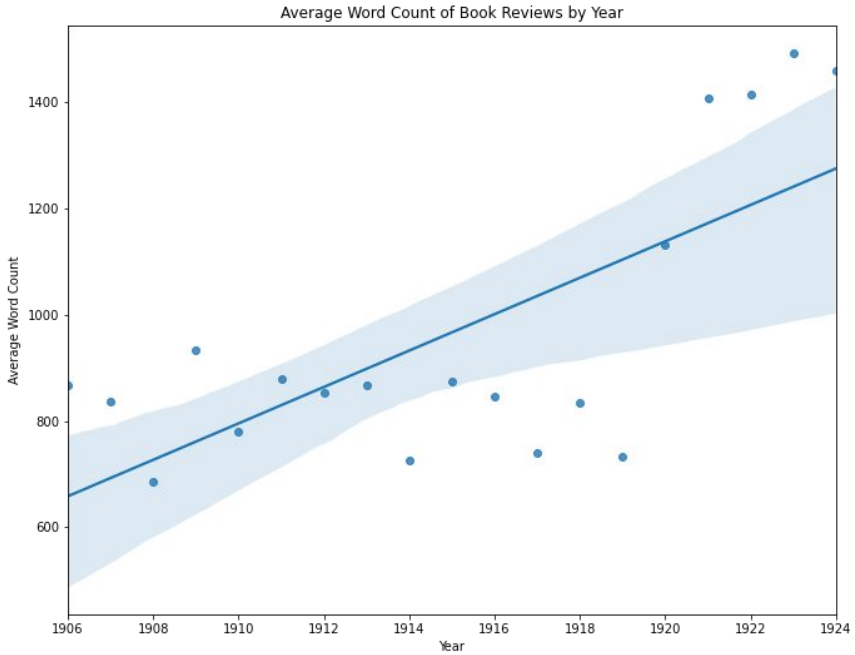
Linear Regression

You have seen this (and done this) before!



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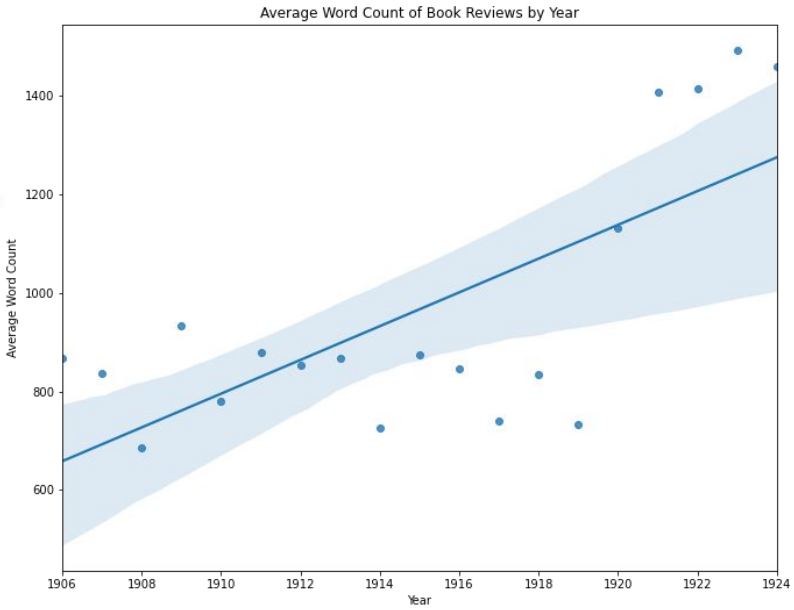
Simple Linear regression prediction function:

$$\hat{y} = b_0 + b_1 x$$

- $\hat{y}$ = the value of the <u>response</u> variable *y*
- x = the explanatory variable (<u>predictor</u>)
- *b0* = the intercept
- *b1* = the slope of the regression line

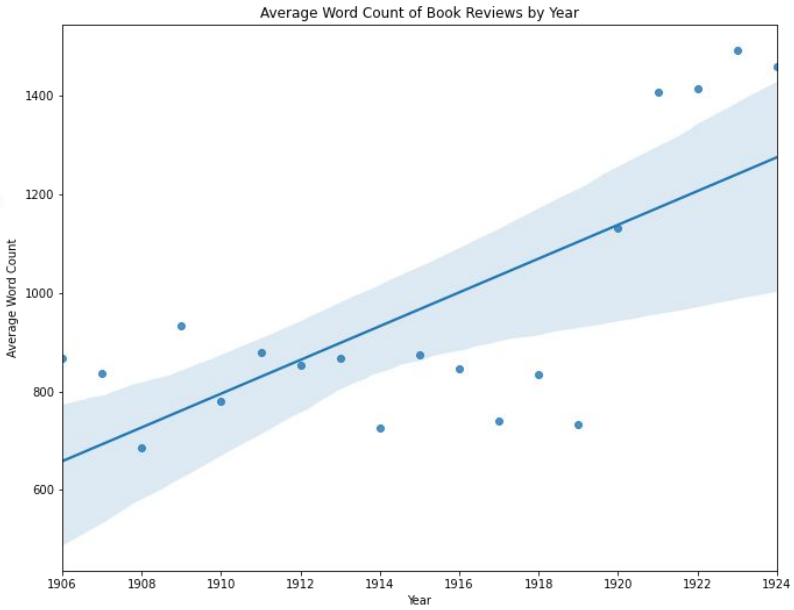


Simple Linear regression prediction function:

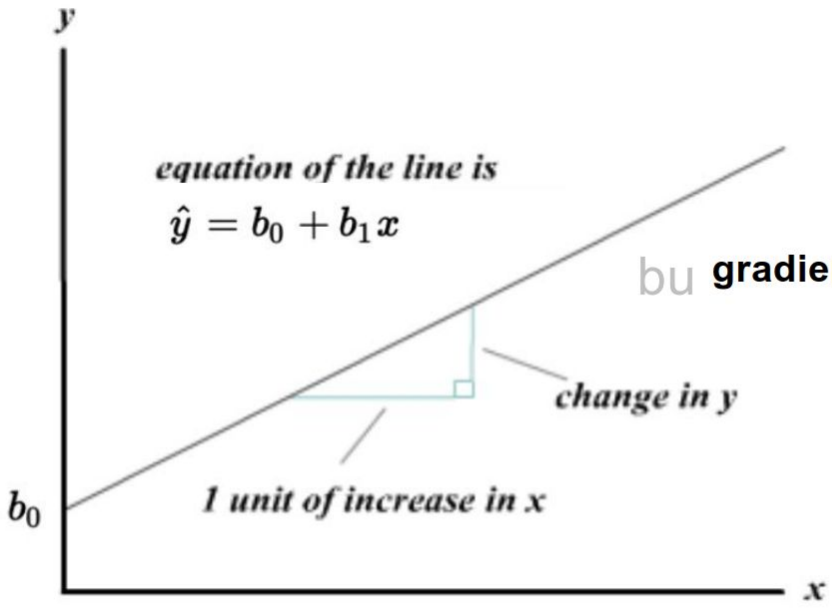
$$\hat{y} = b_0 + b_1 x$$

f(x) = mx + b

- $\hat{y}$ = the value of the <u>response</u> variable *y*
  dependent
- x = the explanatory variable (<u>predictor</u>)
  independent
- *b0* = the intercept
- *b1* = the slope of the regression line



dependent
response variable



$$slope = \frac{\Delta y}{\Delta x}$$

bu **gradient/slope**

predictor variable

independent

$$\hat{y} = b_0 + b_1 x$$

Only a slope or an intercept can not define a line.

Once the intercept and slope are fixed, there can be only one line

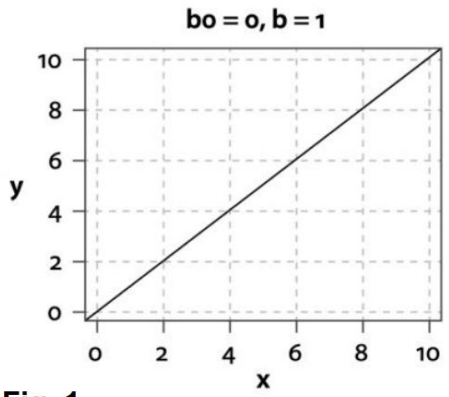


Fig. 1

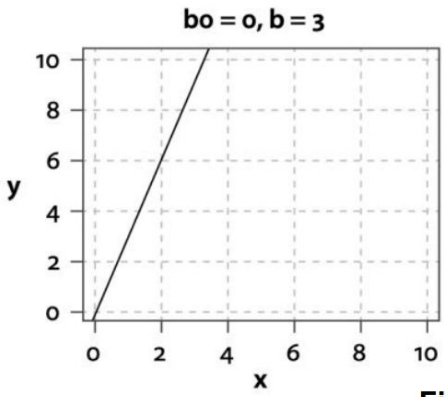


Fig. 2

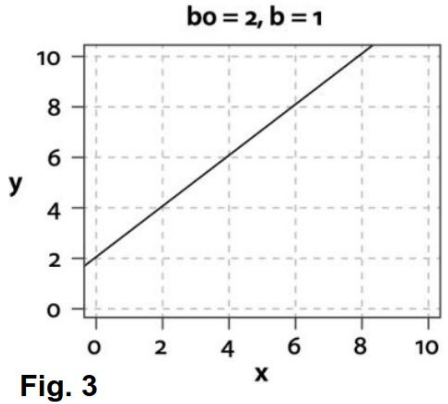


Fig. 3

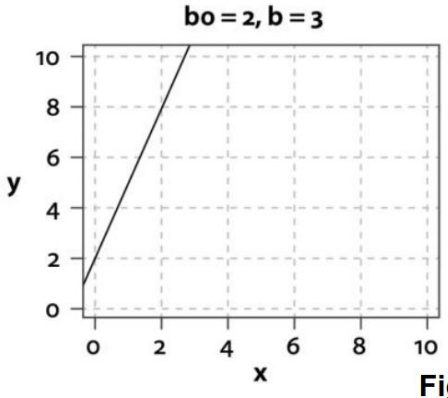
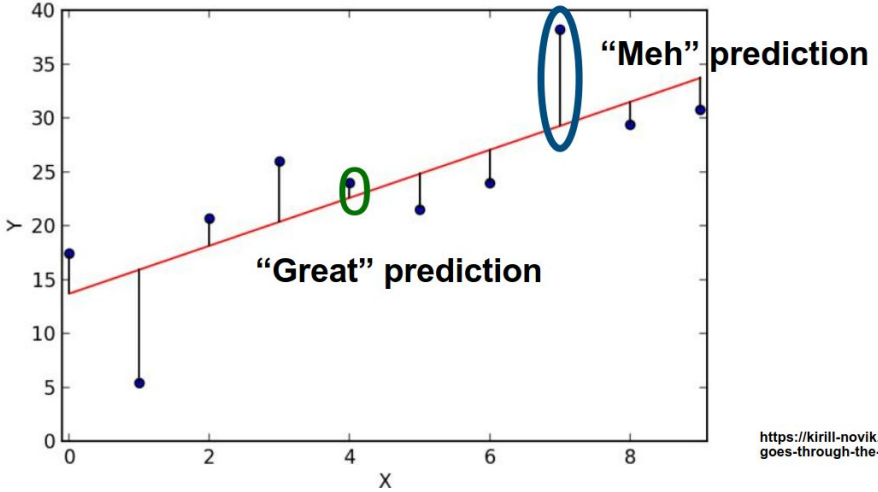


Fig. 4

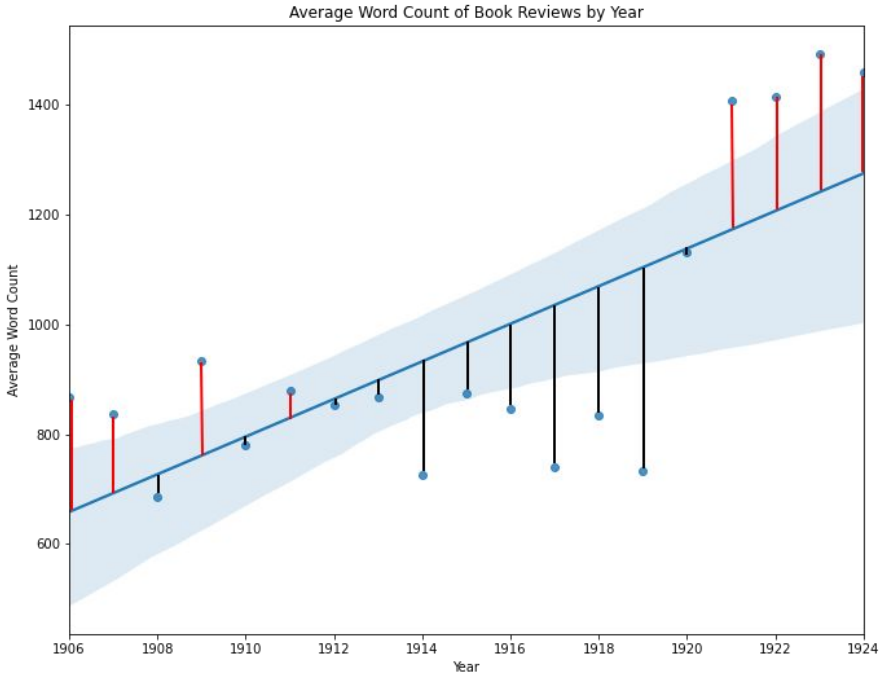
- Residual = the distance between the line and the dots along the line

- Residual: the diference between an observed value, and the fitted value provided by a model



https://kirill-novik.medium.com/linear-regression-line-goes-through-the-mean-point-but-why-93d57914187d

Residuals



Source: Matthew Lavin, [Programming Historian]

How to fit a regression model?

- Objective **minimising the residuals** (= sum of squared error)
- Usually using the **least squares algorithm**
- The **smaller the sum of squared residuals**, the better the model fits the data!

Simple Linear Regression

Try to find values for the slope and intercept that minimize the residual error from the linear model.

Intercept

0 6

Slope

0 3

Show summary(lm(y ~ x))

Link

Simple Linear Regression

Try to find values for the slope and intercept that minimize the residual error from the linear model.

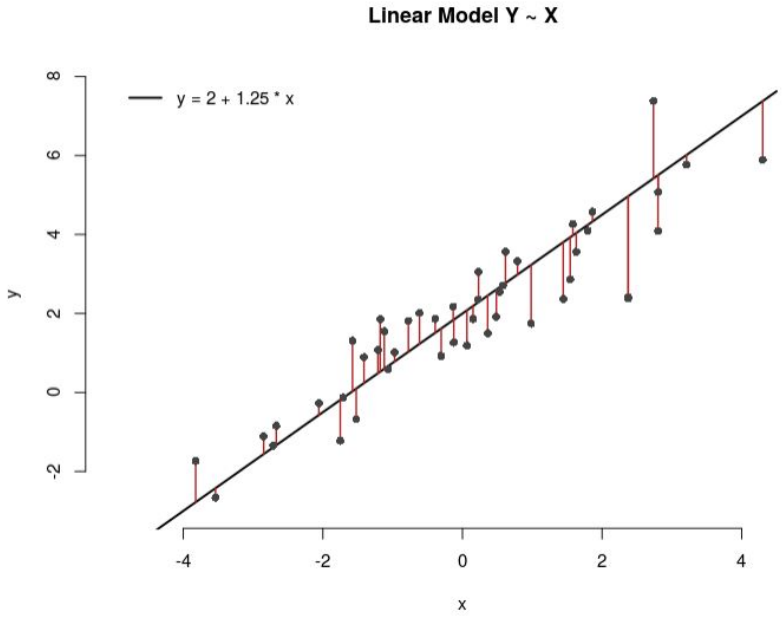
Intercept

-2 2 6

Slope

-1 1.25 3

☑ **Show summary(lm(y ~ x))**



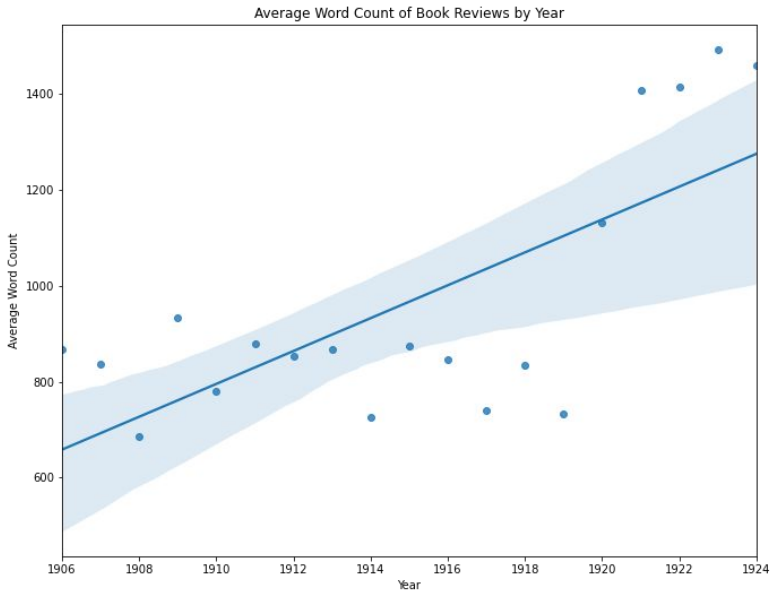
Evaluating Linear Regression Models

After estimating the sample regression line using a set of sample data, we want to know if the estimated line describes the sample data well.

How do we evaluate this?

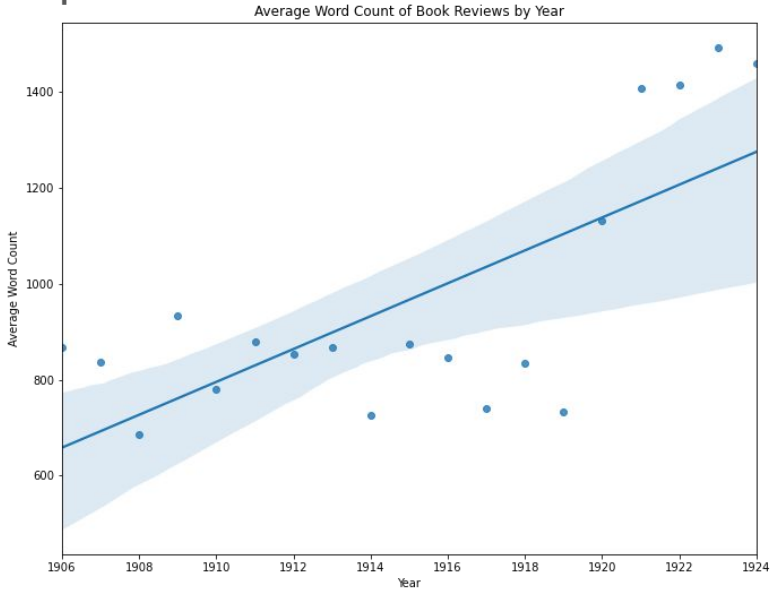
Evaluating Linear Regression Models

We are usually concerned with **how much of the variability in the dependent variable** within the sample is caused by **changes in the independent variable**.



Evaluating Linear Regression Models

The higher the **proportion of the variability in the dependent variable** that is **caused by the independent variable, the better** the estimated line describes our sample data.



R-squared (R^2):

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS}$$

- is the proportion of the regression sum of squares to the total sum of squares;
- with a value range between [0, 1].
- the closer R^2 is to 1, the closer the actual observation points are to the sample line, indicating a higher goodness of fit.

RSS: Residual Sum of Squares (the partial sum of squares of residuals caused by changes in residual terms)
ESS: Explained Sum of Squares (the partial regression sum of squares caused by changes in the independent variable)
TSS: Total Sum of Squares
More about the proof: https://www.youtube.com/watch?v=w2FKXOa0HGA

Intuitively…

If R^2 is close to 1 (or 100%), the **independent** variable X is a good choice for predicting the function that minimizes the residuals of Y.

If R^2 is close to 0, X does not explain Y (**well**).

Linear Regression: Assumptions (1)

- You are predicting (response variable) must be at least **interval-scaled**
  - you are predicting numbers!
  - e.g.: the cost of a house (money) given how large it is (squared meters)

Linear Regression: Assumptions (2)

- Your observations are **independent** from each other
- The relationship between dependent and independent variable is **linear**

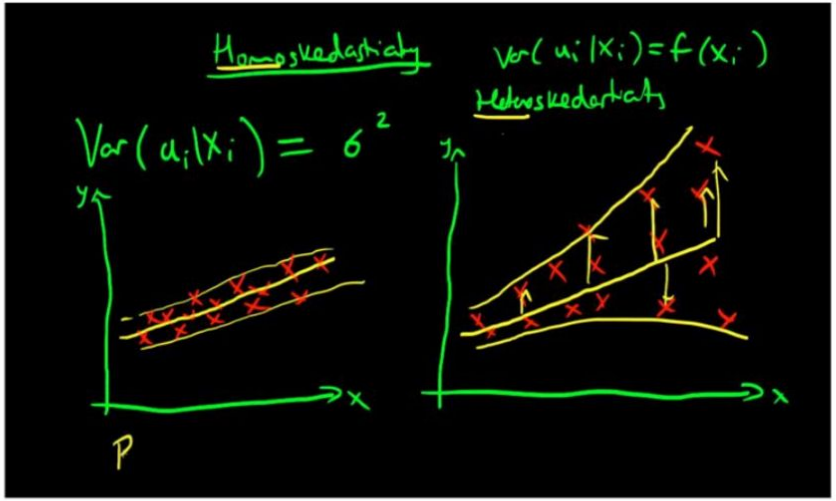
Linear Regression: Assumptions (3)

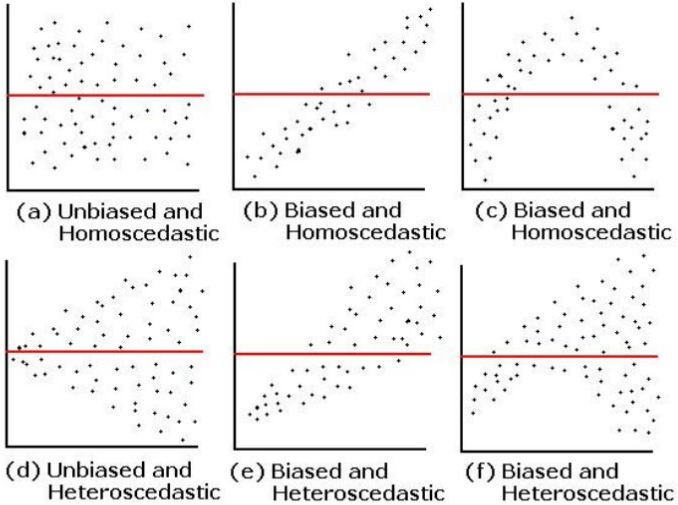
- **Errors** (the residuals) **vary constantly** along the regression line (homoscedasticity)
- The residuals are independent - **no autocorrelation**
- The residuals are **normally distributed**

Linear Regression: Assumptions (3)

Errors are not independent from each other

- Homoscedascity: https://www.youtube.com/watch?v=zRklTsY9w9c





slide from dr. A. Toral Ruiz, right picture from https://baotramduong.medium.com/regression-analysis-homoscedasticity-a76e592f19f2

Link

Multiple Linear Regression Models

If we want to investigate the relationship between **a dependent variable and multiple independent variables**, we use Multiple Linear Regression.

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Intuitively:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n + \epsilon$$

Multiple Linear Regression Models

If we want to investigate the relationship between **a dependent variable and multiple independent variables**, we use Multiple Linear Regression.

Intuitively:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + ... + \beta_n X_n + \epsilon$$

We still look for the best fit function that describes the data!

Multiple Linear Regression Models

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Intuitively:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + ... + \beta_n X_n + \epsilon$$

The relationship is still **linear!**

Multiple Linear Regression Models

Example

Model to predict the level of **narrative complexity** in 19th-century English novels.

Multiple Linear Regression Models

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Model to predict the level of **narrative complexity** in 19th-century English novels.

Dependent variable/Response variable (Y):

Narrative complexity score, e.g. average syntactic tree depth per sentence, or a measure combining sentence length, subordinate clause frequency, lexical diversity ...

Multiple Linear Regression Models

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Multiple Linear Regression Models

Example

Model to predict the level of **narrative complexity** in 19th-century English novels.

Independent variables (X_1, X_2, X_3, …), e.g.:

- Publication year
- Author gender
- Genre (e.g., realist, gothic, sensation)
- Novel length (number of tokens)
- Proportion of direct speech

Multiple Linear Regression Models

Example

Model to predict the level of **narrative complexity** in 19th-century English novels.

Independent variables (X_1, X_2, X_3, …), e.g.:

Narrative complexity = $\beta_0 + \beta_1$(Year) + β_2(Gender) + β_3(Genre) + β_4(Length) + β_5(Dialogue proportion) + ε

Summary

- Machine Learning is a set of methods that help a system to autonomously [ ] and [ ] from given information

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- R2 is a measure of [ ]

Summary

- Machine Learning is a set of methods that help a system to autonomously **learn** and **improve** from given information
- Machine learning algorithms recognize **patterns** in the **training** data
- Three big groups of ML algorithms: **supervised, unsupervised, reinforcement** learning
- Simple Linear Regression works with one **continuous, independent** variable and one **continuous, dependent** variable
- Multiple Linear Regression works with **multiple** continuous, independent variables and **one** continuous, dependent variable
- R2 is a measure of **goodness of fit**